

Hollow Fiber Countercurrent Dialysis for Continuous Buffer Exchange of High-Value Biotherapeutics

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Buffer exchange, desalting, and formulation of high-value biotherapeutics are currently performed using batch diafiltration (DF); however, this type of tangential flow filtration process may be difficult to implement as part of a fully continuous biomanufacturing process. The objective of this study was to explore the potential of using countercurrent dialysis for continuous protein formulation and buffer exchange. Experiments were performed using concentrated solutions of immunoglobulin G (IgG) with commercially available hollow fiber dialyzers having 1.5 and 1.8 m² membrane surface area. More than 99.9% buffer exchange was obtained over a range of conditions, as determined from the removal of a model impurity (vitamin B₁₂). The dialyzers were able to process more than 0.5 kg of IgG per day in an easily scalable low-cost process. In addition, buffer requirements were less than 0.02 L of buffer per gram IgG, which is several times less than that used in current batch DF processes. These results clearly demonstrate the potential of using low-cost hollow fiber dialyzers for buffer exchange and product formulation in continuous bioprocessing. © 2018 American Institute of Chemical Engineers Biotechnol. Prog., 2018

Keywords: countercurrent dialysis, buffer exchange, formulation, diafiltration, antibody, continuous processing

Introduction

Buffer exchange is a critical processing step in the commercial production of all biotherapeutics and other protein-based products.¹ Buffer exchange is used to precondition in-process streams to enhance the performance of subsequent unit operations, for example, desalting before ion exchange chromatography.² More significantly, buffer exchange is required to place the protein product in the desired formulation buffer for long-term storage and delivery.³

The dominant method for performing buffer exchange in large-scale bioprocessing is diafiltration (DF), which involves a simultaneous dilution (buffer addition) and concentration, with the latter performed by tangential flow ultrafiltration in which the liquid phase is removed by pressure-driven filtration through a semipermeable membrane.⁴ Constant volume batch DF is the most common process configuration, with the product solution continuously recirculated through the tangential flow ultrafiltration module while fresh (DF) buffer is added at a rate equal to that at which the filtrate is removed. Commercial systems for batch DF can involve several hundred square meters of membrane area housed in a specially designed stainless steel skid.⁴

Although current bioprocessing operations are almost all performed in batch mode, increasing concerns about cost of goods and manufacturing flexibility has led to growing interest in the potential of continuous bioprocessing.³ These continuous bioprocesses can provide significant reductions in production time and facility size, improved product quality (due to the reduction in residence time throughout the process), and increased productivity.⁵ Pall Corporation has recently introduced the Cadence Inline DF module, the first commercial system designed for continuous buffer exchange.⁶ The DF involves multiple sequential (inline) dilutions and concentration steps, each performed using single pass tangential flow filtration (SPTFF) modules.^{7,8} Thousandfold removal of impurities can be achieved using three stages, each with 10-fold dilution. Although these modules do provide continuous buffer exchange, the system requires somewhat larger buffer volumes to achieve the same level of buffer exchange as a corresponding batch system, and the fluid needs to be managed through a more complicated (and expensive) multihead pumping system.

Nambiar et al.⁹ also used SPTFF modules for continuous DF, but the individual modules (stages) were arranged in a countercurrent configuration to significantly reduce the buffer requirements. Data obtained with a polyclonal immunoglobulin G (IgG) showed 99% removal of a small impurity using a two stage system with similar buffer requirements as a batch

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process. Even higher degrees of impurity removal with lower buffer consumption were predicted for processes employing three or four stages based on classical models for the performance of multistage systems. However, the requirement for multiple pumps, multiple SPTFF modules, and the complex countercurrent staging might limit commercial applications of this technology.

An alternative approach for buffer exchange is to use dialysis in which the buffer components diffuse across a semipermeable membrane that separates the product solution from the dialysate. Dialysis bags or tubes are used extensively for laboratory-scale (batch) buffer exchange,³ with the protein dialyzed against water or an appropriate buffer solution. In addition, hemodialysis (i.e., “blood” dialysis) is one of the largest commercial applications of membrane technology.¹⁰ Current hemodialyzers consist of a parallel array of nearly 10,000 hollow fiber membranes, approximately 200 μm in inner diameter and 40 μm thick.¹¹ Most dialyzers use synthetic polysulfone membranes which are highly retentive to albumin (MW = 67 kDa) but allow rapid removal of small solutes like urea, creatinine, and uric acid.¹¹ Clinical hemodialysis is typically performed with blood flow rates of 300–500 mL/min with dialysate flow rates of 600 mL/min; these dialyzers remove about 90% of the urea in a single pass while also providing significant removal of middle molecular weight proteins like β_2 -microglobulin which is associated with dialysis-related amyloidosis.¹²

Kurnik et al.¹ examined the use of countercurrent dialysis for large-scale buffer exchange in bioprocessing applications as part of a more general study comparing the performance of dialysis, DF, and size exclusion chromatography. The results demonstrated that DF provided the best overall performance, although the analysis was limited to batch processes employing very dilute protein solutions (in this case 1 g/L bovine serum albumin). In addition, the countercurrent dialysis was performed using cuprophane regenerated cellulose membranes that have much less effective transport characteristics than the high performance synthetic polysulfone membranes that now dominate the hemodialysis market.^{10,13}

The objective of this study was to examine the potential of using countercurrent hollow fiber dialyzers for continuous buffer exchange of a highly concentrated protein solution, typical of what would be expected for a continuous bioprocess used for monoclonal antibody (mAb) production. Experiments were performed using off-the-shelf hollow fiber cartridges developed for hemodialysis; these membranes only provide $\leq 90\%$ impurity removal in hemodialysis applications which employ much higher feed flow rates than what were used in this study. The results clearly demonstrate that countercurrent continuous dialysis can readily achieve the high degrees of buffer exchange required for final product formulation using less buffer than a corresponding batch system with relatively simple/straightforward and low-cost operation.

Materials and Methods

All experiments were performed using either Nipro Polynephron[®] or Fresenius Optiflux F180B polysulfone hollow fiber dialyzers with properties summarized in Table 1.

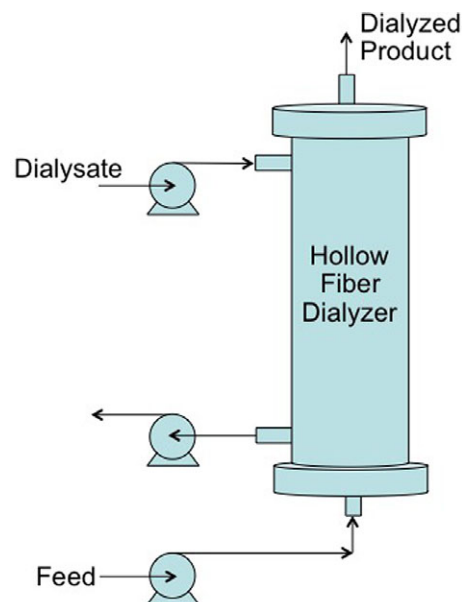


Figure 1. Schematic of continuous countercurrent dialysis system.

A new dialyzer was used for each experiment. The dialyzer was mounted in a vertical orientation with the dialysate buffer introduced into the shell side from the top port (Figure 1). The feed (product solution) was introduced into the lumen-side inlet at the bottom of the dialyzer to achieve countercurrent flow. Feed and buffer flow rates were controlled using Masterflex L/S peristaltic pumps (Cole-Parmer, Vernon Hills, IL) fitted with platinum-cured silicone tubing (Cole-Parmer). The dialysate inlet and outlet flows were controlled using a single pump fitted with two pump-heads; a separate pump was used for the feed inlet. The pumps were calibrated before each experiment by timed collection using a digital balance. Pressures were monitored throughout the experiment using Ashcroft pressure gauges located immediately before and after the inlet/outlet ports. Dialyzers were flushed with buffer solution before the experiment. The experiment was then started with the feed and dialysate flows beginning to fill the dialyzer simultaneously. Experiments were performed at room temperature ($21 \pm 2^\circ\text{C}$) without any external temperature control.

Dialysis was performed using solutions of human serum IgG (NovaBioligics, Oceanside, CA) with vitamin B₁₂ (Cayman Chemicals, Ann Arbor, MI) used as a model impurity. The IgG was dissolved in a 10 mM sodium acetate buffer at pH 5; the same buffer (without any vitamin B₁₂) was used as the dialysate. All solutions were pre-filtered through 0.2 μm mixed cellulose and/or 0.45 μm polyvinylidene fluoride (PVDF) membranes before use to remove any insoluble aggregates. When necessary, the IgG solutions were centrifuged at 4800g before the pre-filtration to remove undissolved (particulate) material.

All experiments were performed using constant feed and buffer flow rates in single pass operation (no recycle of either the IgG or the dialysate). Small samples were obtained periodically from both the dialysate and feed exits for off-line evaluation of the IgG and vitamin B₁₂ concentrations using a

Table 1. Properties of Hollow Fiber Dialyzers

Dialyzer	Manufacturer	Polymer	Membrane Area	Inner Diameter	Fiber Length
Polynephron	Nipro	Polyethersulfone	1.5 m ²	200 μm	0.26 m
Optiflux F180B	Fresenius	Polysulfone	1.8 m ²	185 μm	0.25 m

NanoDrop 2000c Spectrophotometer (Thermo Scientific, Waltham, MA) based on the absorbance at 550 and 280 nm; these wavelengths have strong absorbance for vitamin B₁₂ and IgG, respectively. Appropriate calibration curves were constructed using standard solutions with known concentrations of both components to account for the nonzero absorbance of both species at the two wavelengths.

Results and Analysis

Typical data for the vitamin B₁₂ concentration in the product (exit) stream from a countercurrent dialysis experiment are shown in Figure 2. Data were obtained with the Optiflux F180B dialyzer using a feed flow rate of $q_F = 4.0 \pm 0.3$ mL/min and a dialysate buffer flow rate of $q_D = 9.0 \pm 0.3$ mL/min, corresponding to $\alpha = q_D/q_F = 2.25$. The pressure drop across the dialyzer (feed inlet minus retentate outlet) was <2 kPa (0.3 psi); the very low pressure drop is directly related to the low flow rates used in these experiments compared to the 300–500 mL/min flow rates used in clinical applications of the Optiflux module for hemodialysis. The feed contained IgG and vitamin B₁₂ at concentrations of 100 ± 2 g/L and 3300 ± 100 mg/L, respectively. This system thus processed 0.4 g of IgG per min corresponding to about 0.5 kg/day. This is comparable to the level of protein produced by a 500 L perfusion bioreactor operated continuously using one bioreactor volume exchange per day and a mAb titer of 1–1.5 g/L, which are typical of the values expected for continuous bioprocessing for small-scale manufacturing.⁵ The initial feed volume of 800 mL was fully processed in 200 min. No data were obtained during the first 30 min as it required almost this much time to fully washout the ≈ 100 mL hold-up volume in the dialyzer and tubing.

The vitamin B₁₂ concentration in the final dialyzed product fluctuated between values of 0 and 1.2 ± 0.6 mg/L, corresponding to $99.98 \pm 0.04\%$ buffer exchange; the values plotted in Figure 2 are the average of three consecutive measurements taken at 10-min intervals (plotted at the mean time). The large error bars are due to the very low absorbance of B₁₂ at these low concentrations; these values are just more than the limit of quantification for the spectrophotometric assay. Note that these dialyzers typically provide only 60–90% solute removal in a single pass when used in hemodialysis applications; the

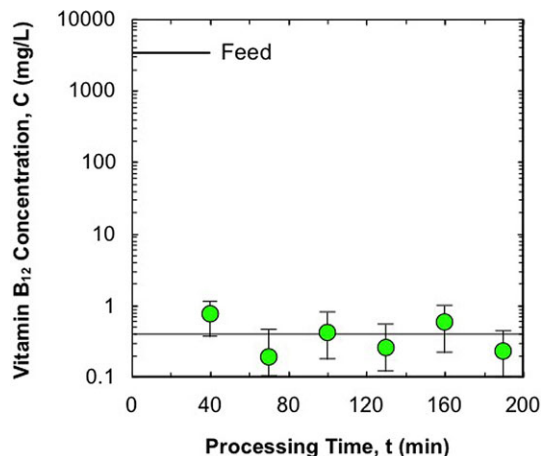


Figure 2. Vitamin B₁₂ concentration in the diafiltered product stream using the Optiflux F180B dialyzer operated with a 4.0 mL/min feed ($\alpha = 2.25$) containing 100 ± 2 g/L IgG and 3300 ± 100 mg/L vitamin B₁₂.

significantly higher level of vitamin B₁₂ removal in this study is due to the much lower flow rates used in our experiments.

The average IgG concentration in the collected product stream was 103 ± 4 g/L (data not shown), which is nearly identical to the feed concentration of 100 ± 2 g/L. The slight increase in IgG concentration was due to a small filtration flow rate across the membrane; the retentate exit flow rate was 3.74 mL/min compared to 3.97 mL/min for the feed. The overall IgG yield was $103 \pm 6\%$ after accounting for the slight concentration of the product stream. There was no evidence of any IgG in the dialysate solution, nor was there any evidence of IgG adsorption to the membrane. This is consistent with the application of these hollow fiber membranes in hemodialysis, in which there is nearly complete recovery of albumin and larger plasma proteins. Note that the buffer exchange in Figure 2 was accomplished using 0.023 L of buffer per g of IgG, which is several times less than that for a conventional batch DF process (0.058 L/g for DF performed at 120 g/L and 0.12 L/g for DF at the more commonly used 60 g/L). This is discussed in more detail subsequently.

Figure 3 shows the degree of vitamin B₁₂ removal, defined as the ratio of the vitamin B₁₂ concentration in the feed to the measured concentration in the buffer exchanged product, as a function of the IgG concentration in the feed solution. Results with both the Optiflux 180B and Nipro Polynephron[®] dialyzers were essentially identical, with $R = C_{\text{Feed}}/C_{\text{exit}} = 1000 \pm 200$ independent of the IgG concentration. This is in sharp contrast to the behavior of a tangential flow DF process in which the filtrate flux decreases significantly with increasing protein concentration.¹⁴ However, increasing the IgG concentration in the feed leads to a corresponding reduction in the amount of buffer required for the buffer exchange, with the run at 120 g/L IgG using less than 0.019 L of buffer per g of IgG processed. Note that the pressure drop through the hollow fiber module remained below 2 kPa (0.3 psi) even at an IgG concentration as high as 120 g/L, consistent with the very low feed flow rates used in these experiments.

The effect of the dialysate (buffer) flow rate on the removal of vitamin B₁₂ is examined in Figure 4 for data obtained with the Optiflux F180B dialyzer using $q_F = 4$ mL/min and IgG feed concentrations between 90 and 120 g/L. In this case, the vitamin B₁₂ concentration in the formulated product was evaluated as the average of samples obtained over the entire length of the continuous dialysis. More than 99.9% buffer exchange ($R > 1000$) was obtained at $q_D = 9$ and 18 mL/min, but the vitamin B₁₂ removal decreased significantly when the dialysate flow rate was reduced to 6 mL/min. The lower rate of buffer exchange is due to the reduction in the concentration driving force between the feed and dialysate solutions (associated with the greater build-up of vitamin B₁₂ in the dialysate at small q_D) in combination with the smaller overall mass-transfer coefficient (associated with the slower flow rate and smaller shear rate in the dialysate region of the dialyzer).

The solid curve in Figure 4 is a model calculation for the fold-removal in a countercurrent dialyzer adapted from Michaels¹⁵:

$$R = \frac{\alpha \exp(\beta) - 1}{\alpha - 1} \quad (1)$$

where

$$\beta = \frac{k_o A}{q_F} \left(1 - \frac{1}{\alpha} \right) \quad (2)$$

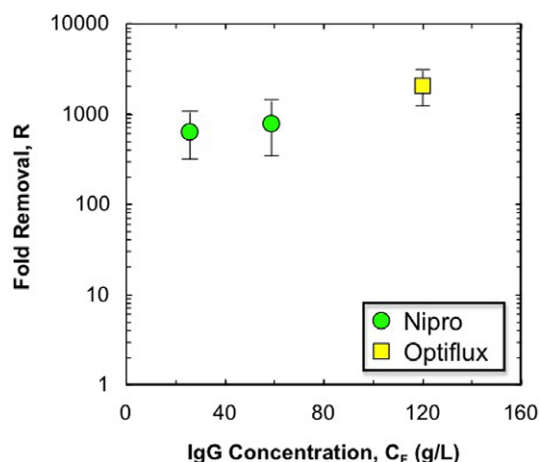


Figure 3. Fold-removal of vitamin B₁₂ as a function of IgG concentration for countercurrent dialysis performed using the Optiflux F180B and Nipro Polynephron[®] dialyzers using a feed flow rate of 4 mL/min and $\alpha = 4.5$.

and $\alpha = q_D/q_F$ with k_oA equal to the product of the overall mass-transfer coefficient and the membrane area. Equation 1 properly describes the reduction in the rate of solute removal with decreasing α using $k_oA = 38$ mL/min (determined by a fit to the eye), although more rigorous comparison between the model and data would require information on the flow rate dependence of the dialysate-side mass-transfer coefficient as discussed by Wickramasinghe et al.¹⁶

In addition to providing high degrees of impurity removal, the countercurrent dialysis system uses significantly less buffer than alternative processes. The volume of buffer used in the countercurrent dialysis per mass of protein is simply:

$$V_D = \frac{\alpha}{C_F} \quad (3)$$

The data in Figure 4 with $\alpha = 2.25$ provided $R = 1000$, which is the typical target for the extent of buffer exchange in the formulation of most biotherapeutics. This corresponds to

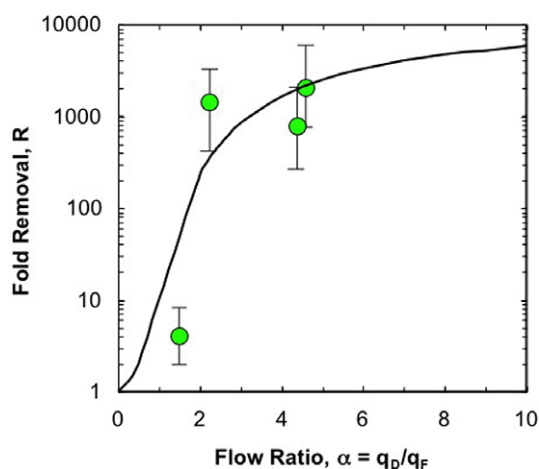


Figure 4. Fold-removal of vitamin B₁₂ as a function of the ratio of the dialysate to feed flow rates (α) for data obtained with the Optiflux F180B dialyzer using a feed flow rate of 4 mL/min. Solid curve is model calculation given by Eq. 1 with $k_oA = 38$ mL/min.

$V_D = 0.019$ L/g assuming a feed concentration of 120 g/L; it is likely that even higher protein concentrations could be used in these dialyzers to further reduce the buffer requirements.

The corresponding expression for the buffer requirement in a conventional batch DF process is⁹:

$$V_D = \frac{N_D}{C_F} = \frac{\ln(R)}{S_o C_F} \quad (4)$$

where N_D is the number of diavolumes and S_o is the observed sieving coefficient for the solute of interest. A batch DF process with $R = 1000$ and $S_o = 1$ (valid for a small impurity) would thus require 0.058 L/g if it were performed with $C_F = 120$ g/L. However, this high a feed concentration would yield a very low filtrate flux; thus, most commercial DF processes are performed with $C_F \approx 60$ g/L giving $V_D = 0.12$ L/g, which is more than six times the amount of buffer required for the countercurrent dialysis system examined in this study.

Recent work by Nambiar et al.,⁹ using the same IgG and vitamin B₁₂ feed solution as that used in this study, demonstrated that it is also possible to perform continuous buffer exchange using a countercurrent staged DF process by using the Pall Inline Concentrator (SPTFF) modules. The buffer required for this type of countercurrent staged system is:

$$V_D = \frac{\alpha}{S_o C_F} \quad (5)$$

where in this case α is equal to the ratio of the DF buffer to feed flow rates. Nambiar et al. (2018) showed theoretically that a three-stage configuration with $\alpha = 9.7$ could provide $R = 1000$ using vitamin B₁₂ as a model impurity, but this would still require four times the amount of buffer than the countercurrent dialysis even if the SPTFF could be performed at the same protein concentration. A summary of the different buffer requirements is provided in Table 2. Note that the use of very high protein concentrations in the batch and three-stage DF systems gives low filtrate flux and large feed-side pressure drops, which would likely create significant challenges in the implementation of a three-stage process for long term continuous operation.

Conclusions

The data obtained in this study clearly demonstrate the feasibility of using countercurrent hollow fiber dialysis for buffer exchange and final formulation of high-value biotherapeutics. The countercurrent dialysis can provide high levels of buffer exchange ($R \geq 1000$) in a fully continuous single-pass process by operating at much lower feed flow rates (<10 mL/min) than that used in hemodialysis ($q_F \geq 300$ mL/min) where the goal is to achieve approximately 70–90% removal of small solutes like urea. In addition, the product is processed with only a single pump pass, compared to the 10–100 pump passes in current batch DF processes; this could lead to significant improvements in product quality, particularly for highly labile products. The results in Figure 2 show stable operation over 3 h, but similar dialyzers have been used for more than 96 h of continuous operation in continuous renal replacement therapy.¹⁷ This long-term operation with minimal fouling is likely due to the absence of any transmembrane filtrate flow; additional studies will be required to determine the potential for even longer term continuous processing using these hollow fiber dialyzers.

Table 2. Comparison of Buffer Requirements for Continuous Countercurrent Dialysis, Batch DF, and Countercurrent Staged DF Processes (With 1000-Fold Buffer Exchange)

	Countercurrent Dialysis ($\alpha = 2.25$)	Batch	Three Stage DF
60 g/L	0.038 L/g	0.12 L/g	0.16 L/g
120 g/L	0.019 L/g	0.058 L/g	0.081 L/g

Although it is always difficult to perform detailed economic analyses of different processes before they have been deployed in actual commercial manufacturing, the results presented in this study suggest that there could be very substantial cost-savings associated with the use of continuous countercurrent dialysis for antibody formulation. First, the countercurrent dialysis system requires severalfold less buffer than either traditional batch or countercurrent staged DF processes. Second, the hollow fiber dialyzers used in this study cost less than \$20 (approximately \$10/m²) for a pre-sterilized cartridge suitable for direct integration in an extracorporeal blood circuit,¹⁸ suggesting that they could also be directly implemented in a continuous downstream biomanufacturing process. If these dialyzers could be used for the 96-h operating time commonly obtained in hemodialysis applications, the cost would be less than \$0.01/g of formulated protein (assuming operation at $q_F = 4$ mL/min and $C_F = 120$ g/L).

Although we operated these dialyzers at low feed flow rates to obtain the high levels of buffer exchange needed in bioprocessing, these systems still processed nearly 0.7 kg of protein per day (based on $q_F = 4$ mL/min and $C_F = 120$ g/L). This is similar to the mass of monoclonal antibody that would be produced by a 500 L perfusion bioreactor operated continuously with one bioreactor volume exchange per day and a titer of ≈ 1 g/L. This suggests that a relatively small number of these hollow fiber dialyzer cartridges operated in parallel could process sufficient material to handle the product generated by continuous operation of a 500–2000 L perfusion bioreactor, which is typical of current targets for large-scale continuous biomanufacturing.¹⁹ In addition, these modules come presterilized and are intended for single-use applications, both of which are highly attractive in bioprocessing. Additional studies will clearly be needed to demonstrate the effectiveness and economics of this type of continuous countercurrent hollow fiber dialysis in commercial bioprocesses.

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